

Project Design Phase

Solution Architecture

Date	15 February 2025
Team ID	
Project Name	Human Development Index (HDI) Predictor
Maximum Marks	4 Marks

Solution Architecture:

The **Solution Architecture** of the **Human Development Index (HDI) Predictor** defines the overall structure and workflow of the AI-powered system. It illustrates how user inputs, data preprocessing, machine learning models, databases, and visualization components work together to provide accurate HDI predictions. The architecture bridges the gap between socio-economic challenges and intelligent technology solutions by ensuring efficient data processing, reliable predictions, and interactive reporting.

Objectives

- Design an AI-powered architecture for accurate Human Development Index (HDI) prediction.
- Define the interaction between the user interface, application layer, machine learning model, database, and visualization components.
- Automate data validation, preprocessing, prediction, and report generation.
- Ensure scalability, reliability, security, and efficient management of socio-economic datasets.
- Provide interactive dashboards and analytical reports to support evidence-based policy and decision-making.

Example - Solution Architecture Diagram:

Figure 1: Solution Architecture of the AI-powered Human Development Index (HDI) Predictor

Human Development Index (HDI) Predictor

Solution Architecture

